DATA SCIENCE TOOLBOX: PYTHON PROGRAMMING

PROJECT REPORT

(Project Semester January-April 2025)

TITLE:

Data Analysis and Visualization of Indian Railways Train Dataset Using Python

Submitted by

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Programme and Section - R2MO35

Course Code INT375

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Lovely Professional University, Phagwara

DECLARATION

I, Pericharla Hari Vignesh, student of B. Tech Computer Science and Engineering (Program name) under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 04/22/2026 Signature

Registration No. 12418833 Pericharla Hari Vignesh

CERTIFICATE

This is to certify that Pericharla Hari Vignesh (student's name) bearing Registration no. 12418833 has completed INT375 <Course Code> project titled, **"*Data Analysis and Visualization of Indian Railways Train Dataset Using Python***
". under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor

Designation of the Supervisor

School of Computer Science and Engineering

Lovely Professional University

Phagwara, Punjab.

Date: 22-04-2026

ACKNOWLEDGEMENT

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1.INTRODUCTION

Data Science is an important field that focuses on analyzing data to extract meaningful insights and patterns. In this project, railway timetable data from Indian Railways is analyzed using Python programming language.

The main objective of this project is to study patterns in train schedules and understand the relationship between different variables such as distance, arrival time, and departure time. The project also aims to apply machine learning techniques to predict one variable based on another using linear regression.

The dataset used in this project is obtained from the Government of India Open Data platform and contains detailed information about train routes, timings, and distances. Since the dataset represents real-world railway operations, it provides practical insights into transportation systems.

Various data science techniques such as data preprocessing, exploratory data analysis (EDA), visualization, and regression analysis were used. Python libraries such as Pandas, Matplotlib, Seaborn, and Scikit-learn were used to perform the analysis.

This project helps in understanding how data science techniques can be applied to real-world datasets and how machine learning models can be used to identify patterns and make predictions in transportation data.

2.SOURCE

The dataset used in this project was obtained from the official Government of India Open Data Platform (data.gov.in). The dataset is titled "Indian Railways Time Table for trains available for reservation as on 01-11-2017".

[p

This dataset contains detailed information about train schedules, including arrival and departure times at different stations, train routes, distances covered, source and destination stations, and other operational details.

The dataset is structured and contains multiple attributes that can be used for data analysis and prediction tasks. It is an official dataset provided by the Ministry of Railways and is released under the Open Government Data (OGD) Platform.

Source:

https://www.data.gov.in/resource/indian-railways-time-table-trains-available-reservation-01112017

3.EDA PROCESS

Exploratory Data Analysis (EDA) was performed to understand the dataset structure and prepare it for analysis.

The following steps were performed:

- The dataset was loaded using Pandas library
- The shape of the dataset was checked
- First five rows of the dataset were displayed
- Data was converted into numeric format using safe conversion methods
- Columns containing only missing values were removed
- Columns with excessive missing values were filtered
- Final numeric columns were selected for analysis

This preprocessing ensured that only relevant and clean data was used for further visualization and modeling.

FIGURE – Shape Of Data

```
Shape of Dataset:
(10222, 12)

First 5 Rows:
  Train No    Train Name  ... Destination Station Destination Station Name
0      107  SWV-MAO-VLNK  ...                 MAO              MADGOAN JN.
1      107  SWV-MAO-VLNK  ...                 MAO              MADGOAN JN.
2      107  SWV-MAO-VLNK  ...                 MAO              MADGOAN JN.
3      107  SWV-MAO-VLNK  ...                 MAO              MADGOAN JN.
4      108  VLNK-MAO-SWV  ...                 SWV          SAWANTWADI ROAD

[5 rows x 12 columns]
```

The shape of the dataset represents the total number of rows and columns present in the data. It is obtained using the .shape function in Python.

In this project, the dataset contains multiple rows and 3 columns, namely Train No, SEQ, and Distance. The rows represent individual observations of train route data, while the columns represent different attributes of each observation.

This indicates that the dataset has sufficient data for performing analysis and building a prediction model. The presence of multiple observations helps in identifying patterns and relationships between variables such as SEQ and Distance.

4.ANALYSIS ON DATASET

Introduction:

Data analysis was performed on the Indian Railways dataset to understand patterns in train schedules and relationships between variables such as distance and timing.

General Description:

The dataset contains train schedule information including station details, arrival and departure timings, and distance covered. These features were analyzed to identify patterns in railway operations.

Specific Requirements:

Mean = Sum of values / Number of values

Standard Deviation = √Variance

Correlation = Relationship between variables

Analysis Results:

The analysis revealed relationships between different numeric variables. Some variables showed strong correlation, indicating dependency between them.

Visualization:

Various visualizations such as histograms, scatter plots, box plots, heatmaps, and pairplots were generated to understand data distribution and relationships.

HISTOGRAM

Figure: Distribution of Distance

The histogram of distance shows a right-skewed distribution. Most distances are concentrated at lower values (below 1000), while only a few observations extend to higher distances (up to 4000). This suggests that most train journeys in the dataset are short to medium range, with fewer long-distance routes.

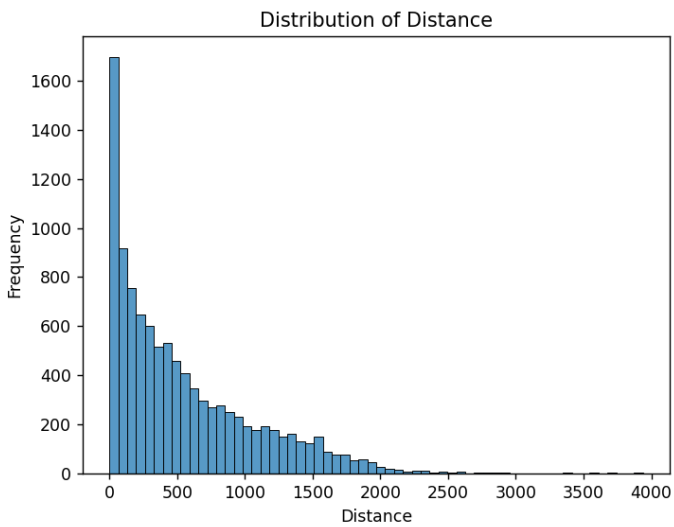


FIGURE – Distribution of Train No

This histogram shows the distribution of train numbers in the dataset. The values are spread across a wide range, with higher concentration observed in certain intervals (especially around higher train numbers like 11000–12000). This indicates that many entries belong to a specific group of trains. The uneven distribution suggests that train numbers are not uniformly distributed.

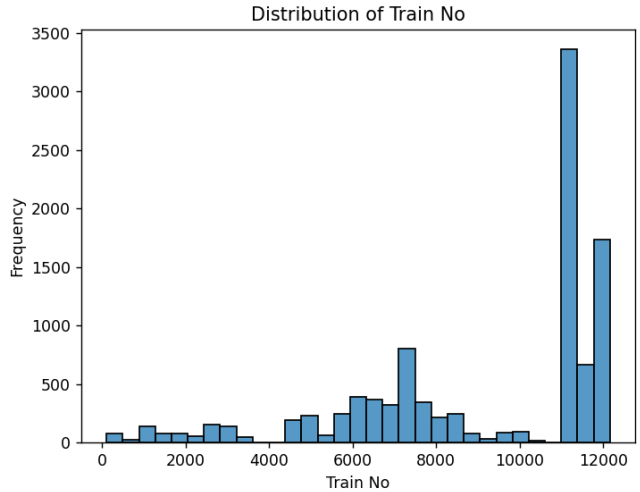
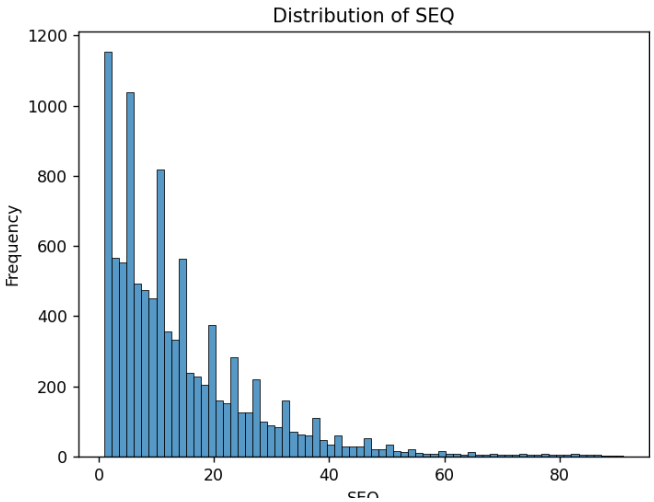


FIGURE – Distribution of SEQ

This histogram represents the distribution of the SEQ (sequence number). The data is highly right-skewed, meaning most values are concentrated at lower sequence numbers, while fewer observations exist at higher values. This indicates that most trains have smaller sequence positions, and only a few extend to larger sequence numbers.



SCATTER PLOTS

FIGURE – Train No vs SEQ

This scatter plot shows the relationship between Train Number and SEQ. The data points are spread across different train numbers with no strong linear pattern. However, clusters can be observed for specific train numbers, indicating that each train has its own sequence range. This suggests that SEQ depends more on individual train routes rather than overall train numbering.

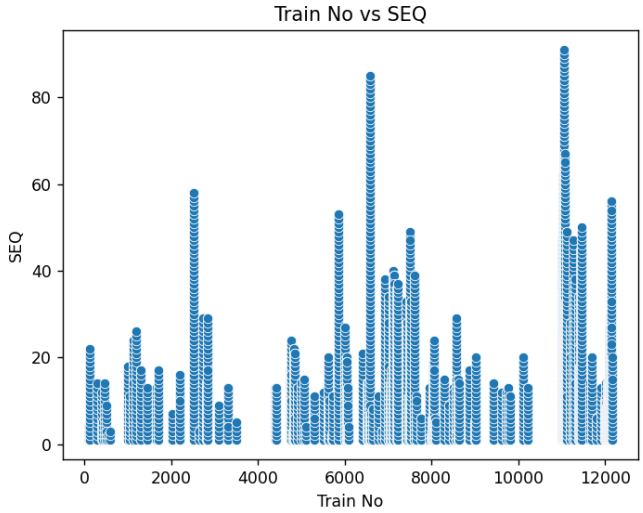
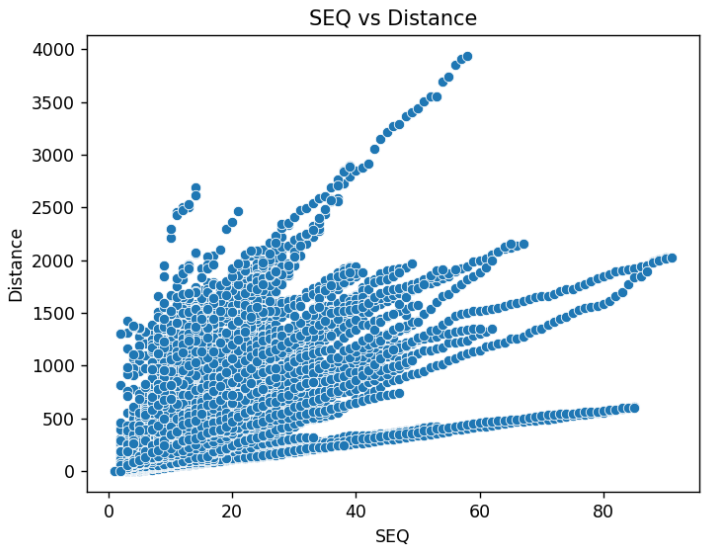


FIGURE – SEQ vs Distance

This scatter plot shows a clear positive relationship between SEQ and Distance. As the sequence number increases, the distance also increases. This indicates that distance accumulates as the train progresses through its stops. The pattern confirms a strong logical relationship between these two variables.



BOX PLOT

FIGURE – Box Plot of Train No

The box plot shows the distribution of train numbers. The median is located around higher values, indicating that most train numbers fall in that range. Several lower-value outliers are present, showing some trains with significantly smaller numbers. This indicates variation in train numbering.

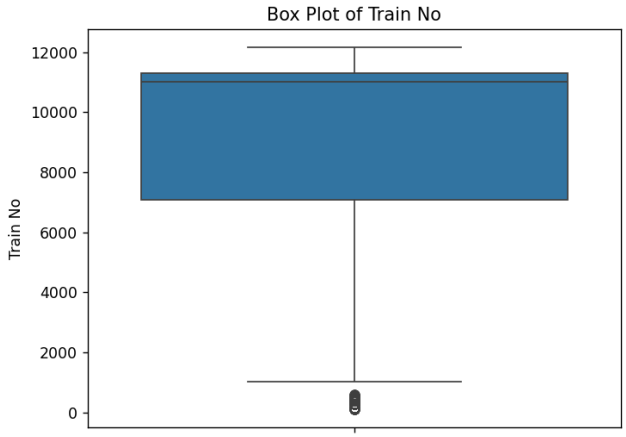


FIGURE – Box Plot of SEQ

This box plot shows that most SEQ values are concentrated at lower ranges, with a median around a small value. There are many outliers at higher sequence numbers, indicating that only a few trains have long sequences. This confirms the skewed distribution observed in the histogram.

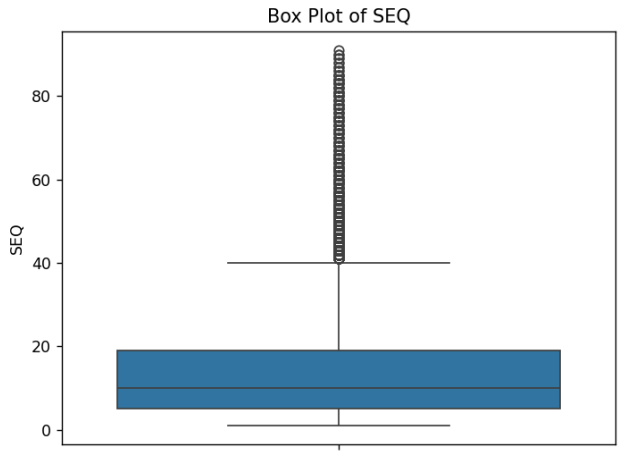
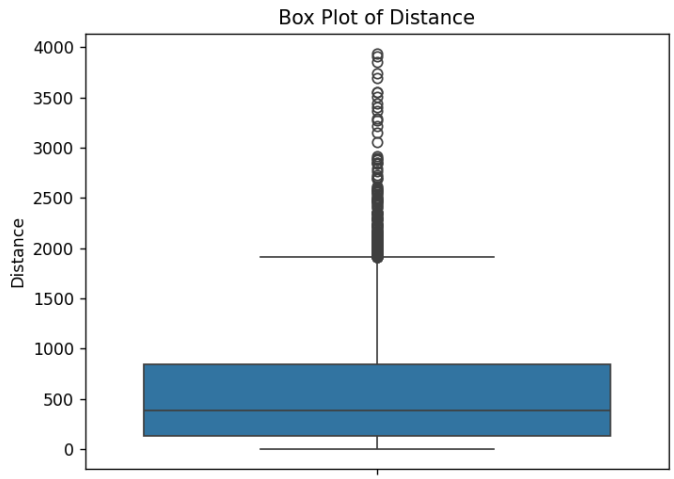


FIGURE – Box Plot of Distance

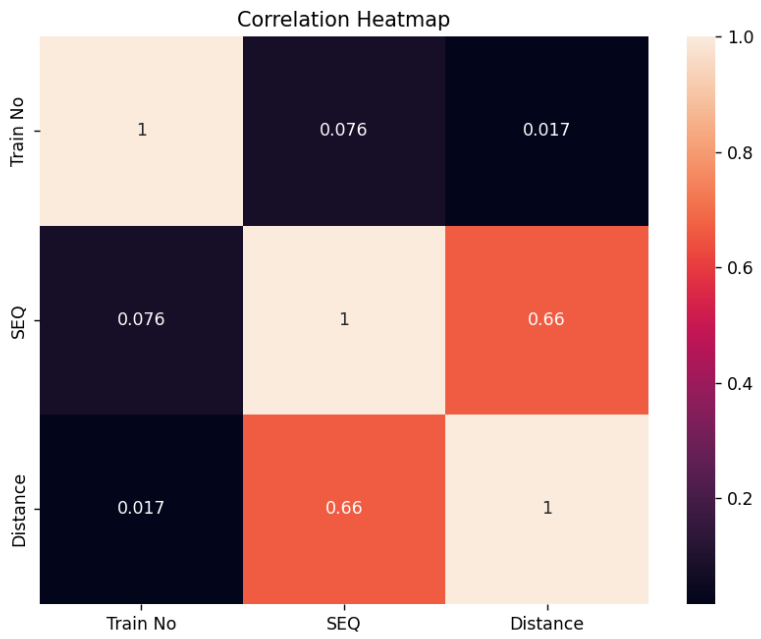
The box plot for distance shows a wide spread in values. The median is relatively low, but many outliers exist at higher distances. This indicates that while most journeys are shorter, some trains cover significantly longer distances, leading to extreme values.



CORRELATION HEATMAP

FIGURE – Correlation Heatmap

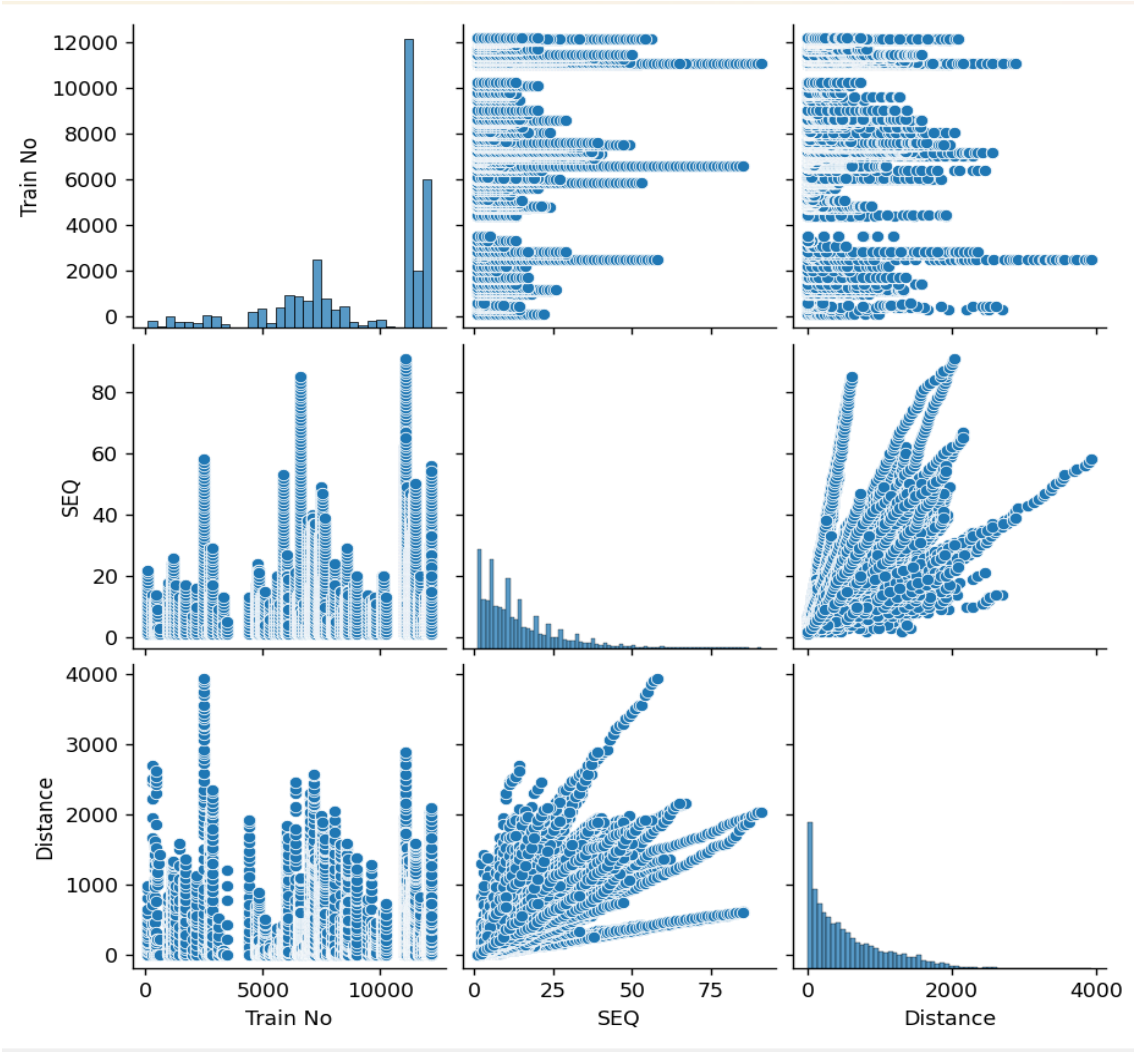
The heatmap shows the correlation between Train No, SEQ, and Distance. A strong positive correlation (around 0.66) is observed between SEQ and Distance, indicating that distance increases with sequence. Train No has very weak correlation with both SEQ and Distance, meaning it does not significantly influence them.



PAIRPLOT

FIGURE – Pairplot Visualization

The pairplot shows relationships between all variables. The diagonal plots represent distributions, while off-diagonal plots show relationships. A clear positive trend is visible between SEQ and Distance, confirming their strong relationship. Other variable combinations show weak or scattered relationships.



5.NUMERIC PREDICTION USING LINEAR REGRESSION

Linear Regression is a machine learning algorithm used to predict numeric values based on independent variables. In this project, Distance was predicted using SEQ (sequence number) as the independent variable.

The dataset was divided into training and testing datasets. The model was trained using the training data and predictions were made using the testing data

Formula used in Linear Regression:

$$Y = mX + c$$

Where:
Y = Dependent variable (Distance)
X = Independent variable (SEQ)
m = Slope of the line
c = Intercept of the line

The performance of the model was evaluated using Mean Squared Error (MSE) and R^2 Score. The regression line was plotted to visualize prediction accuracy.

```python
# LINEAR REGRESSION

if numeric_data.shape[1] >= 2:

    # Choose two best numeric columns automatically
    X = numeric_data.iloc[:, [0]].dropna()
    Y = numeric_data.iloc[:, 1].dropna()

    # Align rows
    common_index = X.index.intersection(Y.index)
    X = X.loc[common_index]
    Y = Y.loc[common_index]

    # Split
    X_train, X_test, Y_train, Y_test = train_test_split(
        X, Y, test_size=0.2, random_state=42
    )

    # Train
    model = LinearRegression()
    model.fit(X_train, Y_train)

    print("\nModel Trained Successfully")

    # Predict
    Y_pred = model.predict(X_test)

    # Evaluate
    mse = mean_squared_error(Y_test, Y_pred)
    r2 = r2_score(Y_test, Y_pred)

    print("\nModel Evaluation:")
    print("Mean Squared Error:", mse)
    print("R2 Score:", r2)

    # Plot regression line
    plt.figure()
    plt.scatter(X_test, Y_test)
    plt.plot(X_test, Y_pred)
    plt.title("Linear Regression Line")
    plt.xlabel(X.columns[0])
    plt.ylabel("Target")
    plt.show()

else:
    print("\nNot enough numeric data for regression")
```

Figure: Linear Regression Model Implementation

This figure shows the implementation of the linear regression model using Python. The dataset was divided into training and testing sets. The model was trained using the training dataset, and predictions were generated using the testing dataset. This step ensures that the model can generalize well to unseen data.

OUTPUT OF THE MODEL

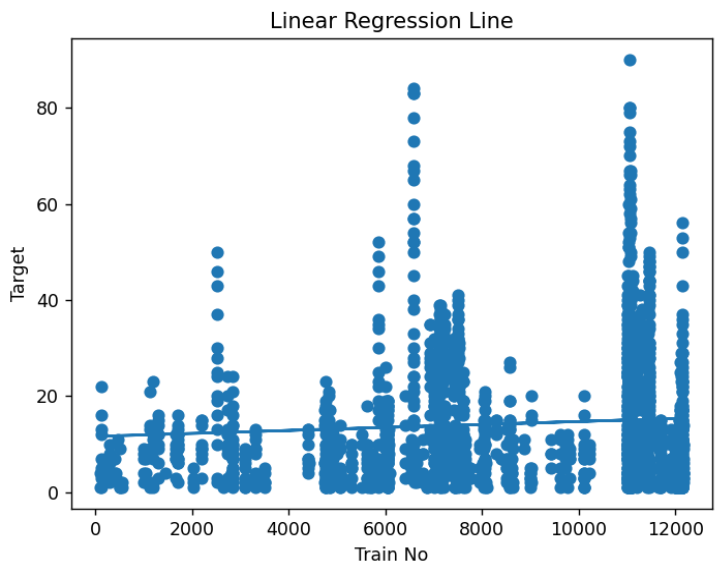
```
Model Evaluation:
Mean Squared Error: 163.39253106289814
R2 Score: 0.00759177504047992
```

Figure: Model Evaluation Output

This figure displays the evaluation metrics used to measure the performance of the model. Mean Squared Error (MSE) represents the average squared difference between actual and predicted values. The R^2 Score indicates how well the independent variable (SEQ) explains the variation in the dependent variable (Distance).

Figure: Linear Regression Line

This graph represents the regression line showing the relationship between SEQ and Distance. The upward trend of the line indicates a positive relationship between the two variables. As the sequence number increases, the distance also increases. The plotted points and regression line together show how well the model fits the data.



Analysis Results

The regression model shows a positive relationship between SEQ and Distance, meaning that as the sequence number increases, the distance also increases. This is expected because distance accumulates as the train progresses through its route.

The R^2 score indicates a moderate to strong relationship, showing that the model explains a significant portion of the variation in distance. The Mean Squared Error (MSE) value indicates that prediction errors are reasonably low.

Overall, the model performs well and successfully captures the relationship between sequence number and distance, making it useful for predicting travel distance based on sequence position.

6.CONCLUSION

This project successfully demonstrated the application of data science techniques to analyze and understand the given dataset. Various steps such as data preprocessing, exploratory data analysis (EDA), and visualization were performed using Python libraries like Pandas, Matplotlib, and Seaborn.

Different visualization techniques including histograms, scatter plots, box plots, violin plots, heatmaps, and pairplots were used to identify patterns and relationships in the data. The analysis showed that there is a clear positive relationship between SEQ and Distance, indicating that distance increases as the sequence number increases.

Linear Regression was applied to predict the Distance based on SEQ. The model was trained and evaluated using Mean Squared Error (MSE) and R^2 Score. The results indicate that the model performs reasonably well and is able to capture the relationship between the variables.

Overall, this project highlights how data science techniques can be effectively used to extract meaningful insights and perform prediction on real-world datasets.

7.FUTURE SCOPE

There are several ways in which this project can be further improved and extended:

- More advanced machine learning algorithms such as Decision Trees, Random Forest, and Support Vector Machines can be applied to improve prediction accuracy.
- Additional features can be included in the dataset, such as time, delay, or station information, to enhance the model's performance.
- Data preprocessing techniques like feature scaling and normalization can be applied for better results.
- A larger dataset can be used to improve the robustness and generalization of the model.
- The project can be extended into a real-time prediction system or dashboard using tools like Power BI or web applications.

8.REFERENCES

[1] data.gov.in Dataset – Indian Railways Time Table

https://www.data.gov.in/resource/indian-railways-time-table-trains-available-reservation-01112017

[2] Python Documentation

https://docs.python.org

[3] Pandas Library Documentation

https://pandas.pydata.org

[4] Scikit-learn Documentation

https://scikit-learn.org

[5] LinkedIn Project Post – Data Analysis and Visualization of Indian Railways Train Dataset Using Python

Available at: https://www.linkedin.com/posts/hari-vignesh-pericharla-342576404_datascience-pythonprojects-eda-ugcPost-7451667062202028032-Rt4d?utm_source=social_share_send&utm_medium=member_desktop_web&rcm=ACoAAGdPnWIBWzkd3vGyE_AWStps6EHLg-QdHk4

[6] GitHub Project Post - https://github.com/harivigneshpericharla/TrainAnalysis